

## C++ Test

Test Reference No: PLMAcademy-Sep2025-STMT-N-General-Batch

Total Marks : 30

Test Duration : 1 hour

### Instructions to Candidate

1. All questions are compulsory/mandatory.
2. Do not write anything on the question paper.  
Write everything on the answer sheet provided.
3. This is a closed book test.
4. There is no negative marking.
5. Wherever there is ambiguity, make suitable assumptions and proceed.  
Document the assumptions in the answer.
6. Assume C++ 11 language standard.
7. Assume usage of Microsoft Visual Studio development platform.

| Question No | Question                                                                                                                                                                                    | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1           | Write the C++ declaration for a reference to a pointer to an integer.                                                                                                                       | 1     |
| 2           | Write the C++ declaration for a pointer to a constant integer.                                                                                                                              | 1     |
| 3           | Explain the use of typedef with a suitable example. Give a minimal example.                                                                                                                 | 1     |
| 4           | Explain call by value, call by address and call by reference with a suitable minimal example.                                                                                               | 3     |
| 5           | For the following code, explain the symbol table.<br><br><pre>void g( );<br/><br/>void f ( )<br/>{<br/>}<br/><br/>int main( ){<br/>    f();<br/>    g();<br/><br/>    return 0;<br/>}</pre> | 2     |
| 6           | What is <i>name mangling</i> and why it is used in C++?                                                                                                                                     | 2     |

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| 7  | <p>For the following code write a function pointer pointing to the function f and call that function. Make suitable assumptions.</p> <pre> int f ( int i ) {     return 2 * i; }  int main ( ) {     _____      _____      return 0; } </pre>                    | 1 |
| 8  | Write a C++ code for a parameterized default constructor.                                                                                                                                                                                                        | 1 |
| 9  | In C++, how is encapsulation done. Show with a suitable example.                                                                                                                                                                                                 | 2 |
| 10 | <p>What will be the preprocessed output for the following code?</p> <p><b>File a.h</b></p> <pre> #ifdef A #define VAL 10 #else #define VAL 20 #endif </pre> <p><b>File a.cpp</b></p> <pre> #include "a.h"  int main() {     int i = VAL;      return 0; } </pre> | 1 |
| 11 | <p>Ramesh is given the following files:</p> <p><b>File a.h</b></p> <pre> void f(); </pre>                                                                                                                                                                        | 2 |

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|    | <p><b>File a.cpp</b></p> <pre>#include "a.h"  int main() {     f();     return 0; }</pre> <p>Ramesh tries to compile and link the above code, using the command</p> <pre>cl.exe a.cpp</pre> <p>Will the code compile and link?<br/>If no, what will be the error? Why is the error coming?</p>                                                                          |   |
| 12 | Explain the difference between function declaration and function definition.                                                                                                                                                                                                                                                                                            | 1 |
| 13 | <p>Suresh wants to write a declaration for an array of pointers. Demonstrate to him how he can write the code for an array of integer pointers and store the address of variable i and j in the array of integer pointers.</p> <pre>int main ( ); {     int i = 10;     int j = 20;      _____; // Declare here an array of pointers to integer.      return 0; }</pre> | 2 |
| 14 | Explain the concept of l-value and r-value in C++ with a suitable example.                                                                                                                                                                                                                                                                                              | 2 |
| 15 | <p>What will be the output of the following code?</p> <pre>#include &lt;iostream&gt; using namespace std;  int main() {</pre>                                                                                                                                                                                                                                           | 1 |

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|    | <pre> int i = 0;  if( i = 0 ){     cout &lt;&lt; "Value of i is zero" &lt;&lt; endl; } else {     cout &lt;&lt; "Value of i is non-zero" &lt;&lt; endl; }  return 0; } </pre>                                                                                                                |   |
| 16 | <p>What will be the output of the following code:</p> <pre> #include &lt;iostream&gt; using namespace std;  int main() {     int i = 10;      while( i &gt; 0 ){         cout &lt;&lt; i &lt;&lt; endl;         i = i - 2;     }     return 0; } </pre>                                      | 1 |
| 17 | <p>Explain function overloading, with a suitable example.<br/>Give minimal example.</p>                                                                                                                                                                                                      | 2 |
| 18 | <p>Write a function f which dynamically allocates memory for an array of 5 doubles, and returns the address of the array.</p> <p>Also, write the function main, where a call is made to the function f. In main, ensure that the dynamically allocated memory from function f, is freed.</p> | 2 |
| 19 | <p>Will the following code compile?<br/>If so, what will be the output of the program.<br/>If not, what will be the error.</p> <pre> #include &lt;iostream&gt; using namespace std;  class X { </pre>                                                                                        | 1 |

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|    | <pre> X() {     cout &lt;&lt; "X default constructor called" &lt;&lt; endl; } };  int main ( ) {     X x1;      return 0; } </pre>                                                         |   |
| 20 | <p>In the following code is the variable p created in stack memory area or in heap memory area?</p> <pre> int main() {     int *pi = new int[2];      delete [] pi;     return 0; } </pre> | 1 |